

Brac University
Department of Computer Science and Engineering
Spring 2025

Name:
Student ID:
Section:

10 Marks
25 Minutes

1. Let $f(x) = x^2 \sin(x)$, where $x \in \{\pi/3, \pi/2\}$.

- a. What will be the degree of the polynomial for above nodes? [1]
- b. Construct the Vandermonde Matrix, V for above nodes. [2]
- c. Using Lagrange Interpolation, find the interpolating polynomial. [3]
- d. Using the Vandermonde matrix found in answer (b) check if the interpolating polynomial is the same as the polynomial in answer (c) [4]

a) degree of the polynomial will be 1.

b)

$$V = \begin{bmatrix} 1 & \pi/3 \\ 1 & \pi/2 \end{bmatrix}$$

c)

$$P_1(x) = L_0(x) f(x_0) + L_1(x) f(x_1)$$

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - \pi/2}{\pi/3 - \pi/2} = \frac{x - \pi/2}{\frac{2\pi - 3\pi}{6}} = -\frac{6(x - \pi/2)}{\pi}$$

$$L_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - \pi/3}{\pi/2 - \pi/3} = \frac{x - \pi/3}{\frac{3\pi - 2\pi}{6}} = \frac{6(x - \pi/3)}{\pi}$$

$$P_1(x) = -\frac{6(x - \pi/2)}{\pi} \times \left(\frac{\pi}{3}\right)^2 \sin\left(\frac{\pi}{3}\right) + \frac{6(x - \pi/3)}{\pi} \times \left(\frac{\pi}{2}\right)^2 \sin\left(\frac{\pi}{2}\right)$$

$$= (-6n + 3\pi) \frac{\pi}{9} \sin \pi/3 + (6n - 2\pi) \frac{\pi}{4} \sin \pi/2$$

$$= -1.813799364 n + 2.849109379$$

$$+ 4.71238898 n - 4.934802201$$

$$= 2.898589616 n - 2.085692822$$

$$\underline{\text{all}} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & \pi/3 \\ 1 & \pi/2 \end{bmatrix}^{-1} \begin{bmatrix} (\pi/3)^2 \sin(\pi/3) \\ (\pi/2)^2 \sin(\pi/2) \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -2 \\ -1.909859317 & 1.909859317 \end{bmatrix} \begin{bmatrix} 0.949703126 \\ 2.4674011 \end{bmatrix}$$

$$= \begin{bmatrix} -2.085692822 \\ 2.898589616 \end{bmatrix}$$

$$= 2.898589616 n - 2.085692822$$

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1. Let $f(x) = x^2 \sin(x)$, where $x \in \{\pi/2, \pi/3\}$.

- What will be the degree of the polynomial for above nodes? [1]
- Construct the Vandermonde Matrix, V for above nodes. [2]
- Using Lagrange Interpolation, find the interpolating polynomial. [3]
- Using the Vandermonde matrix found in answer (b) check if the interpolating polynomial is the same as the polynomial in answer (c) [4]

a) degree of the polynomial will be 1

b)
$$V = \begin{bmatrix} 1 & \pi/2 \\ 1 & \pi/3 \end{bmatrix}$$

c)
$$P_1(x) = L_1(x)f(x_1) + L_2(x)f(x_2)$$

$$L_1(x) = \frac{x - x_2}{x_1 - x_2} = \frac{x - \pi/3}{\pi/2 - \pi/3} = \frac{x - \pi/3}{\frac{2\pi - \pi}{2}} = \frac{x - \pi/3}{\pi/2} = \frac{2(x - \pi/3)}{\pi}$$

$$L_2(x) = \frac{x - x_1}{x_2 - x_1} = \frac{x - \pi/2}{\pi/3 - \pi/2} = \frac{x - \pi/2}{\frac{2\pi - 3\pi}{2}} = \frac{x - \pi/2}{-\pi/2} = -\frac{2(x - \pi/2)}{\pi}$$

$$P_1(x) = \frac{2(x - \pi/3)}{\pi} \left(\frac{\pi}{2}\right)^2 \sin(\pi/2) - \frac{2(x - \pi/2)}{\pi} \left(\frac{\pi}{3}\right)^2 \sin(\pi/3)$$

$$= (6n - 2\pi) \frac{\pi}{4} \sin \frac{\pi}{2} + (-6n + 3\pi) \frac{\pi}{9} \sin \frac{\pi}{3}$$

$$= 2.898589616 n - 2.085692822$$

$$d) \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & \pi/2 \\ 1 & \pi/3 \end{bmatrix}^{-1} \begin{bmatrix} (\pi/2)^2 \sin(\pi/2) \\ (\pi/3)^2 \sin(\pi/3) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -2 & 3 \\ 1.909859317 & -1.909859317 \end{bmatrix} \begin{bmatrix} 2.4674011 \\ 0.949703126 \end{bmatrix}$$

$$= 2.898589616 n - 2.085692822$$